

Interpret division remainders

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Calculate $6,745 \div 32$ and interpret the answer for 32-seat coaches. Copy or complete the first step.

2. Calculate $5,126 \div 24$ and write the remainder as a fraction.

3. Share 3,850 stickers equally among 16 classes. How many each and how many left?

4. Miro says: Miro always rounds a quotient with a remainder up. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Calculate $6,745 \div 32$ and interpret the answer for 32-seat coaches. Copy or complete the first step.
Answer 210 remainder 25, so 211 coaches are needed.
2. Calculate $5,126 \div 24$ and write the remainder as a fraction.
Answer 213 remainder 14, or $213 \frac{7}{12}$.
3. Share 3,850 stickers equally among 16 classes. How many each and how many left?
Answer 240 each with 10 left.
4. Miro says: Miro always rounds a quotient with a remainder up. Is this correct? Explain.
Answer The context decides whether to round up, round down, keep the remainder or express it as a fraction.

Interpret division remainders

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Calculate $6,745 \div 32$ and interpret the answer for 32-seat coaches.

6. Check this result using an inverse, estimate or known fact: Share 3,850 stickers equally among 16 classes. How many each and how many left?

2. Calculate $5,126 \div 24$ and write the remainder as a fraction.

7. A 2,500 ml drink fills 180 ml cups. How many full cups?

3. Share 3,850 stickers equally among 16 classes. How many each and how many left?

8. Identify and correct this misconception: Miro always rounds a quotient with a remainder up.

4. Solve this independently and show every stage: Calculate $6,745 \div 32$ and interpret the answer for 32-seat coaches.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Calculate $5,126 \div 24$ and write the remainder as a fraction.

10. Write and solve a similar question to: Share 3,850 stickers equally among 16 classes. How many each and how many left?

Teacher answers and checking prompts

1. Calculate $6,745 \div 32$ and interpret the answer for 32-seat coaches.
Answer 210 remainder 25, so 211 coaches are needed.
2. Calculate $5,126 \div 24$ and write the remainder as a fraction.
Answer 213 remainder 14, or $213 \frac{7}{12}$.
3. Share 3,850 stickers equally among 16 classes. How many each and how many left?
Answer 240 each with 10 left.
4. Solve this independently and show every stage: Calculate $6,745 \div 32$ and interpret the answer for 32-seat coaches.
Answer 210 remainder 25, so 211 coaches are needed.
5. Use a second method or representation for: Calculate $5,126 \div 24$ and write the remainder as a fraction.
Answer Expected result: 213 remainder 14, or $213 \frac{7}{12}$. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Share 3,850 stickers equally among 16 classes. How many each and how many left?
Answer Expected result: 240 each with 10 left. A valid check must be shown.
7. A 2,500 ml drink fills 180 ml cups. How many full cups?
Answer 13 full cups with 160 ml left.
8. Identify and correct this misconception: Miro always rounds a quotient with a remainder up.
Answer The context decides whether to round up, round down, keep the remainder or express it as a fraction.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Share 3,850 stickers equally among 16 classes. How many each and how many left?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Interpret division remainders

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. A 2,500 ml drink fills 180 ml cups. How many full cups?

6. Create a new example that tests the same idea as: Share 3,850 stickers equally among 16 classes. How many each and how many left?

2. Prove the answer to this question, rather than only calculating it: Calculate $6,745 \div 32$ and interpret the answer for 32-seat coaches.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Calculate $5,126 \div 24$ and write the remainder as a fraction.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 240 each with 10 left.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro always rounds a quotient with a remainder up.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. A 2,500 ml drink fills 180 ml cups. How many full cups?
Answer 13 full cups with 160 ml left.
2. Prove the answer to this question, rather than only calculating it: Calculate $6,745 \div 32$ and interpret the answer for 32-seat coaches.
Answer Expected result: 210 remainder 25, so 211 coaches are needed. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Calculate $5,126 \div 24$ and write the remainder as a fraction.
Answer Expected result: 213 remainder 14, or $213 \frac{7}{12}$. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 240 each with 10 left.
Answer One valid reconstruction is: Share 3,850 stickers equally among 16 classes. How many each and how many left? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro always rounds a quotient with a remainder up.
Answer The context decides whether to round up, round down, keep the remainder or express it as a fraction.
6. Create a new example that tests the same idea as: Share 3,850 stickers equally among 16 classes. How many each and how many left?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.